MARK SCHEME · TEACHER COPY

10BR · Chapter 2: Data Transmission

Cambridge IGCSE Computer Science 0478

Paper 1 Computer Systems

Total marks: 33

Official Cambridge mark scheme extracts for each question in the matching exam paper, with question numbers changed to match the exam. Source of each question is shown above its rows.

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 Total: 33 marks

Question 1 · May/June 2024 Paper 12, original Q2 (a),(b),(c) [7 marks]

Question	Answer	Marks
1(a)	**D**	**1**
1(b)	Packet	**1**
1(c)(i)	Data is **encrypted** and **decrypted** using the **same** key (1 mark) Any **three** from: • Plain text is encrypted into cipher text // cipher text is decrypted into plain text • **Data** is encrypted using an **algorithm** … • … that uses a key • The **key** can be generated using an **algorithm** • The **key** is **transmitted** to the receiver	**4**
1(c)(ii)	Any **one** from: • To help keep the data secure • To make the data meaningless	**1**

Question 2 · May/June 2024 Paper 12, original Q6 [5 marks]

Question	Answer	Marks
2	**One** mark for each correct method.	**5**

error detection method	statement
parity (check/bit/byte/block)	An odd or even process can be used.
checksum	A value is calculated from the data using an algorithm. This happens before and after the data is transmitted.
echo check	A copy of the data is sent back to the sender by the receiver.
automatic repeat query/request // ARQ	Acknowledgement and timeout are used.

	check digit	A value is appended to data that has been calculated using the data. This value is checked on data entry.

Question 3 · May/June 2026 Paper 11, original Q4 (b),(c) [6 marks]

Question	Answer	Marks	Guidance
3 ·	Examiners must **consistently** apply the requirements of the mark scheme and any rules for what to do when candidates have not followed instructions correctly.		
3(a)	**Four** from: • (Serial) transmits the data one bit at a time • … down a single wire/channel/line • (Half-duplex) transmits the data (between the computer and the mouse) in both directions/ways … • … but only one direction/way at a time	**4**	MP1 – BOD bit by bit MP2 –cable/path is TV
3(b)	**Two** from: • Cannot plug the cable in the wrong way round • **Different** data transmission speeds are supported • High/adequate speed of data transmission • Universal connection/standard • Connection is **backwards** compatible • It will power the mouse // It will charge the mouse (if it has wireless capabilities) • The mouse will be **automatically** detected // Mouse drivers will be **automatically** downloaded // plug and play • Unlikely to have interference/crosstalk	**2**	Mark whole answer. Do not accept 'can be used over long distance'; this is not relevant to the context. Do not accept reference to skewing/errors as this is given in the question. MP3 – BOD fast/faster transmission. Fast/faster alone is NE. MP8 – Mouse is automatically connected is TV

Question 4 · May/June 2026 Paper 12, original Q3 (e) [5 marks]

Question	Answer	Marks	Guidance

Total: 33 marks

4	Examiners must award marks **positively**. This means that: • marks are not deducted for errors or omissions • marks are awarded for correct/valid answers as defined in the mark scheme and also for **other valid answers** which go beyond the scope of the syllabus and mark scheme referring to your team leader as appropriate • the quality of spelling, punctuation and grammar are judged **only** when the mark scheme indicates that these features are **specifically assessed** in the question. However, the meaning of all responses must be unambiguous.		
4(a)(i)	• Header • Trailer	**2**	Mark first answer given. BOD head for header and trail for trailer.
4(a)(ii)	**Three** from: • It examines/analyses/searches packet to find the **destination IP** address // It reads the **destination IP** address for a packet • It **sends/transmits/forwards** packet towards/to its **destination** • It **determines/assigns/controls/selects** the **route/path** each packet takes • … which is the **fastest/most efficient/shortest** route/path // which is the route with the **least traffic** • Uses/updates a routing table • … to keep track of the most efficient routes/paths • Checks/updates the packets hop count/time to live/TTL • … and disregards packets when they have reached their hop count	**3**	Do not accept generic statement like 'each packet takes a different path' if there is no reference to the router determining/assigning the path. MP1 - Accept MAC address in place of IP address. With a LAN this would be used. MP2 – must have reference to the router forwarding the packet to its destination. It isn't enough to just say router determines path for the packet to reach its destination. This would get MP3 but not MP2 as well. MP3 and MP4 can be given for a single statement e.g. It determines the fastest route for a packet. This gets MP3 for determining the route and MP4 for it being the fastest. MP4 – best path is TV

Question 5 · May/June 2026 Paper 12, original Q4 (a),(b),(c) [10 marks]

Question	Answer	Marks	Guidance
5 ·	Examiners must **consistently** apply the requirements of the mark scheme and any rules for what to do when candidates have not followed instructions correctly.		
5(a)	**One** mark for each correct parity bit. • 1 • 1 • 1	**3**	Mark first answer given.

5(b)(i)	**Two** from: • A transposition error has occurred // Bits have been interchanged/changed **places** // **Even** number of bits has flipped/changed // bits are out of order // doesn't record the location of bits • … so count/total of the 1s/0s in the byte is still **odd/the same** // still has an **odd** number of 1s/0s • … so it will still meet the parity set	**2**	BOD data in place of bits MP1 – Even number of errors is TV MP2 – number of 1s is correct is NE
5(b)(ii)	**Two** from: • Parity **block** check • **Even** parity (byte) check • Checksum • Echo check • Automatic repeat query/request // ARQ	**2**	Mark first answer given. Do not accept check digit. This is for data entry and not transmission. Context is about transmission. Accept error detection methods not listed in the syllabus e.g. cyclic redundancy check. If unsure, refer to TL.
5(c)(i)	**One** from: • Symmetric // synchronous • Asymmetric // asynchronous	**1**	Mark first answer given. Accept by example e.g. public and private key encryption Do not allow protocols that use encryption e.g. SSL. This is not a method of encryption.
5(c)(ii)	**One** from: • A string of **characters/numbers** used to encrypt data • Used by an **algorithm/cipher** (to encrypt/decrypt the data)	**1**	MP2 – Accept 'It is an algorithm/cipher'
5(c)(iii)	Cipher/cypher text	**1**	Mark first answer given. BOD 'cipher' alone or 'cipher data'